

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – SHORT ANSWER TEST

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO5 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 3 – Short Answer Test
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
Student Name:	A. Likhithasri	Reg. No.:	192511003
Section / Batch:	1	Date:	02/09/2026

Code of Conduct

I A. Likhithasri (192511003) (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: A. Likhithasri

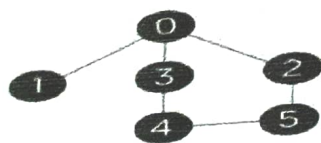
Instructions

- This test contains 10 short answer test questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Annexure A - Short Answer Test

- Perform the following operations in the Singly Linked List for the following input.
Input: 22-> 77-> 55-> 88->
 - Insert a new-node 110 at the end of the list.
 - Insert a new-node 101 at the position 4 of the list.
 - Delete a node at the beginning of the single linked list.
 - Delete a node at the position 3 of the single linked list.
- Rotations in each step of insertion. 1,2,3,4,5,6,7,8,9,10 (0, 4) Push (90), 5) Push (36), 6) Push (11), 7) Push (88), 8) Pop (0, 9) Pop (0). Draw the diagram of the stack and illustrate the above operations and identify where the top is?
- Convert the following infix into a postfix expression. $\rightarrow B - (C/D + (E \% F * G) / H) * J$
- Evaluate the given postfix expression. $53+62/*35*+$
- Construct a B-Tree of Order 3 by inserting numbers 34,26,45,12,87,56,78,22,23,24,65,85,95.
- Construct a 2-3-4 Tree by inserting the following sequence of numbers 45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48.
- Build a max heap for the following numbers - 20, 35, 23, 22, 4, 45, 21, 5, 42 and 19. At each stage of heap tree construction, ensure heap properties are satisfied.
- Perform the operations onto splay tree (draw after each operation): insert (3), insert (9), insert (12), insert (55), insert (1), insert (3), delete (3), delete (9), delete (60), insert (3), search (12).
- Describe the algorithm to sort the elements 170, 45, 75, 90, 802, 24, 2, 66 using Radix sort. Explain with each phase?
- Consider the following graph & construct BFS.



Graph with six nodes

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30	Demonstrates complete and accurate understanding of concepts	Good understanding with minor gaps	Basic understanding	Limited understanding
Accuracy of Answer	25	Answers are completely correct and relevant	Mostly correct with minor errors	Partially correct	Mostly incorrect
Problem-Solving & Reasoning	20	Provides logical reasoning and appropriate steps	Good reasoning with minor gaps	Basic reasoning	Lacks logical reasoning
Clarity & Relevance	15	Answers are precise, clear, concise, and directly address the question	Mostly clear and relevant	Somewhat unclear or incomplete	Irrelevant or unclear answers
Time Management & Completions	10	Completes all questions accurately within the time	Completes most questions on time	Requires additional time	Unable to complete the test

SFaculty In-charge	Course Coordinator	Program Director
Signature:	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

C05 Assessment Tool-03

- Short answer test

Name:

Reg No:

1. Singly linked list operations.

Initial list : $22 \rightarrow 77 \rightarrow 55 \rightarrow 88$

(i) Insert 110 at end.

$22 \rightarrow 77 \rightarrow 55 \rightarrow 88 \rightarrow 110$

(ii) Insert 101 at position 4 $\rightarrow$

$22 \rightarrow 77 \rightarrow 55 \rightarrow 101 \rightarrow 88 \rightarrow 110$

(iii) Delete at beginning

$77 \rightarrow 55 \rightarrow 101 \rightarrow 88 \rightarrow 110$

(iv) Delete at position 3

$77 \rightarrow 55 \rightarrow 88 \rightarrow 110$

Stack operations:

Sequence : $\text{Push}(90), \text{Push}(36), \text{Push}(11), \text{Push}(88), \text{pop}(), \text{pop}()$

• After $\text{Push}(90)$: $\text{Top} = 90 \rightarrow [90]$

• $\text{push}(36)$: $\text{Top} = 36 \rightarrow [90, 36]$

• $\text{push}(11)$: $\text{Top} = 11 \rightarrow [90, 36, 11]$

• $\text{push}(88)$: $\text{Top} = 88 \rightarrow [90, 36, 11, 88]$

• $\text{pop}()$: removes 88 $\rightarrow [90, 36, 11]$ ($\text{Top} = 11$)

• $\text{pop}()$: removes 11 $\rightarrow [90, 36]$ ($\text{Top} = 36$)

3. Infix $\rightarrow$ postfix

Expression : $B \rightarrow (C(D + (E \cdot F) G) / H) * J$

Apply operator precedence and parentheses:

postfix : $BCD|EF \cdot |G * H| + J * -$

4. Evaluate postfix:

Expression : $53 + 62 / + 35 *$

Stepwise

- $53 + \rightarrow 8$
- $62 / \rightarrow 3$
- $8 * 3 \rightarrow 24$
- $5 + 3 \rightarrow 15$
- $24 + 15 \rightarrow 39$

Final result = 39.

5.

B-Tree of order 3

Insert sequence : 34, 26, 45, 12, 87, 56, 78, 22, 23, 24, 65, 85, 95.

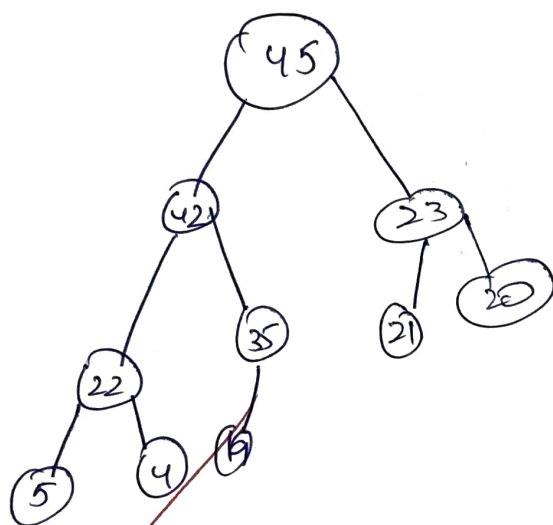
- Nodes split when more than 2 keys.
- Root eventually contains middle values (e.g., 34, 56, 78)
- children hold ranges like $[12, 22, 23, 24, 26]$, $[45]$, $[65, 85, 87, 95]$

2-3-4 Tree construction.

- Insert sequence : 45, 36, 46, 23, 89, 115, 98, 39, 41, 56, 69, 48.
- splits occur when a node has 4 keys.
- Root holds medians (e.g. : 45, 76, 98)
- children grouped by ranges : left [23, 36, 39, 41], middle [48, 56, 69], right [89, 115]

7) Max Heap construction:

Numbers : 20, 35, 23, 22, ⁴¹45, 21, 5, 42, 19.



largest element (45) at root, heap property satisfied.

8) splay Tree operations:

Insertions : 3, 9, 12, 55, 1, 3 → each splayed to root.
Deletions : remove 3, 9, 60 (60 not found)

Insert 3 again then search 12 $\rightarrow$ 12 splayed
to root.

Final root = 12, with balanced subtrees around it.

9. Radix sort

Numbers : 170, 45, 75, 90, 802, 24, 66.

- phase 1 (units) : sort by last digit $\rightarrow$ [802, 2, 24, 45, 75, 66, 170, 90]
- phase 2 (tens) : sort by tens $\rightarrow$ [802, 2, 24, 45, 66, 75, 170, 90]
- phase 3 (hundreds) : sort by hundreds $\rightarrow$ [2, 24, 45, 66, 75, 90, 170, 802]

Final sorted list = [2, 24, 45, 66, 75, 90, 170, 802]

10. BFS Traversal:

Graph with 6 nodes (0-5)

start at 0 $\rightarrow$ visit neighbours level by level

Order: 0, 1, 2, 3, 4, 5.

BFS explores breadth first, ensuring, shortest path discovery.